

Recall — three-dimensional shapes

Naming the solids, counting their faces, edges and vertices, and a pattern that links the three.

LESSON 175

1 × 40 MIN

MATEMATIKA 6, TAKRORLASH

S7 8.4 3-D SHAPES

LESSON CLOCK

10'

12'

12'

6'

Naming the solids	10 min
Faces, edges and vertices	12 min
The pattern $F + V - E$	12 min
Prisms and pyramids in general	6 min

LEARNING OBJECTIVES

- Name the common solids and say which have curved surfaces.
- Count the faces, edges and vertices of a prism and a pyramid.
- Use the pattern $F + V - E = 2$ as a check.
- Give the counts for a prism or pyramid on an n -sided base.

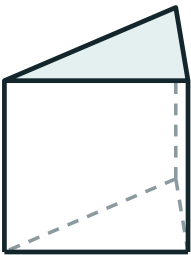
TEXTBOOK

National	pp. 498–501
Cambridge	Stage 7, pp. 84–88
Workbook	Exercise 8.4

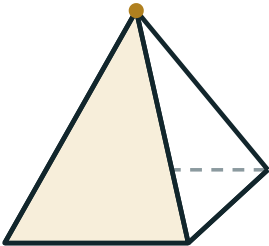
01

Explanation

Naming the solids



prism



pyramid

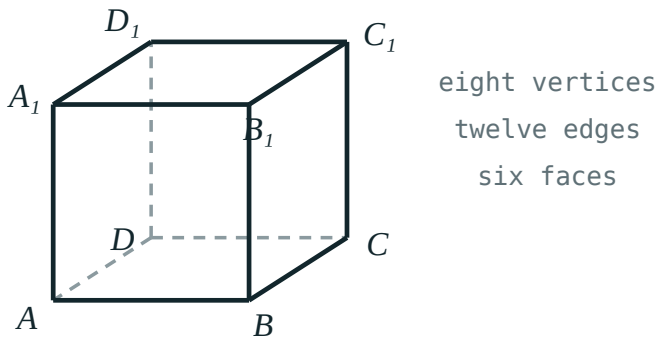
A prism and a pyramid — the two families that most solids belong to

SOLID	FLAT FACES	CURVED SURFACES	RECOGNISE IT BY
cube and cuboid	6	0	six rectangles
prism	$n + 2$	0	two identical ends, joined by rectangles
pyramid	$n + 1$	0	a base and triangles meeting at an apex
cylinder	2	1	two circles and a tube
cone	1	1	a circle and an apex
sphere	0	1	every point the same distance from the centre

A PRISM IS NAMED BY ITS CROSS-SECTION

Cut a prism anywhere across and the face is the same shape every time — triangular, pentagonal, hexagonal. A cuboid is simply a rectangular prism.

Faces, edges and vertices



Six faces, twelve edges, eight vertices

SOLID	FACES F	EDGES E	VERTICES V
cube or cuboid	6	12	8
triangular prism	5	9	6
pentagonal prism	7	15	10
square pyramid	5	8	5
triangular pyramid	4	6	4
hexagonal pyramid	7	12	7

COUNT THE HIDDEN ONES TOO

A drawing shows at most three faces of a cube, but there are six. Turning the solid — or the drawing — in the mind is the skill this recall is really about.

The pattern $F + V - E$

Add the faces and the vertices, subtract the edges, and every solid with flat faces gives the same answer.

$F + V - E = 2$

SOLID	$F + V$	E	$F + V - E$
cube	14	12	2
triangular prism	11	9	2
square pyramid	10	8	2
hexagonal prism	20	18	2
triangular pyramid	8	6	2

USE IT TO CHECK A COUNT, NOT TO MAKE ONE

If your three numbers do not give 2, one of them is wrong — and it is almost always the edges, because some of them are hidden. The rule does not apply to a cylinder, a cone or a sphere: their surfaces are not made of flat polygons.

Prisms and pyramids in general

SOLID ON AN n -SIDED BASE	FACES	EDGES	VERTICES
prism	$n + 2$	$3n$	$2n$
pyramid	$n + 1$	$2n$	$n + 1$

BOTH FORMULAS PASS THE CHECK

For the prism, $(n + 2) + 2n - 3n = 2$. For the pyramid, $(n + 1) + (n + 1) - 2n = 2$. With $n = 6$ they give 8, 18, 12 and 7, 12, 7 — the last two rows of the previous table.

02 Worked examples

EXAMPLE 1 Count the faces, edges and vertices of a triangular prism, and check the pattern.

1

Two triangular ends and three rectangles: $F = 5$.

2

Three edges at each end and three joining them: $E = 9$.

3

Three vertices at each end: $V = 6$; $5 + 6 - 9 = 2$ ✓

ANSWER

5 faces, 9 edges, 6 vertices

EXAMPLE 2 A pyramid has a hexagonal base. Find its faces, edges and vertices.

- ① The base and six triangles: $F = 7$.
- ② Six base edges and six sloping ones: $E = 12$.
- ③ Six base vertices and the apex: $V = 7$; $7 + 7 - 12 = 2$ ✓

ANSWER

7 faces, 12 edges, 7 vertices

EXAMPLE 3 A pupil counts 8 faces, 12 vertices and 16 edges for a hexagonal prism. Find the mistake.

- ① $8 + 12 - 16 = 4$, not 2.
So a count is wrong.
- ② A hexagonal prism has $3n = 18$ edges.
Six top, six bottom, six upright.
- ③ The two missing edges are the hidden uprights.
 $8 + 12 - 18 = 2$ ✓

ANSWER

The edges: there are 18, not 16

03 Interactive model

Pass round a real prism and a real pyramid while the counting is done; the hidden edges stop being hidden the moment the solid can be turned over.

Counting on a solid

A cube has how many edges?

A triangular prism has how many faces?

A square pyramid has how many vertices?

$F + V - E$ for any solid with flat faces is:

0

1

2

it varies

A cylinder has how many curved surfaces?

0

1

2

3

A cone has how many flat faces?

0

1

2

3

A prism on an n -sided base has how many edges?

$2n$

$3n$

$n + 2$

$n + 1$

A pyramid on an n -sided base has how many faces?

n

$n + 1$

$n + 2$

$2n$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	ЎЗБЕКЧА	РУССКИЙ
Solid	Jism	Тело
Prism	Prizma	Призма
Pyramid	Piramida	Пирамида
Cylinder	Silindr	Цилиндр
Cone	Konus	Конус
Sphere	Shar	Шар
Apex	Uchi (cho'qqi)	Вершина (апекс)
Curved surface	Egri sirt	Криволинейная поверхность

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A prism is named by:

its height

its cross-section

its colour

its volume

A sphere has how many flat faces?

0

1

2

6

A triangular pyramid has how many faces?

3

4

5

6

If your counts give $F + V - E = 4$ then:

the solid is unusual

a count is wrong

the rule fails

nothing

Euler's pattern does not apply to:

a prism

a pyramid

a cylinder

a cube

A cuboid is:

a pyramid

a rectangular prism

a cylinder

a cone

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. The faces of a cube

ANSWER 6

2. The edges of a cube

ANSWER 12

3. The vertices of a cube

ANSWER 8

4. The faces of a triangular prism

ANSWER 5

5. The edges of a triangular prism

ANSWER 9

6. The vertices of a triangular prism

ANSWER 6

7. The flat faces of a sphere

ANSWER 0

Medium · the standard question

7 PROBLEMS

1. Faces, edges and vertices of a square pyramid

ANSWER 5, 8, 5

2. Faces, edges and vertices of a triangular pyramid

ANSWER 4, 6, 4

3. Faces, edges and vertices of a pentagonal prism

ANSWER 7, 15, 10

4. $F + V - E$ for a cube

ANSWER 2

5. $F + V - E$ for a square pyramid

ANSWER 2

6. The curved surfaces of a cylinder

ANSWER 1

7. The apexes of a cone

ANSWER 1

Hard · stretch and reason

7 PROBLEMS

1. Faces, edges and vertices of a hexagonal prism

ANSWER 8, 18, 12

2. Faces, edges and vertices of a hexagonal pyramid

ANSWER 7, 12, 7

3. The faces of a prism on an n -sided base

ANSWER $n + 2$

4. Its edges

ANSWER $3n$

5. Its vertices

ANSWER $2n$

6. The edges of a pyramid on an n -sided base

ANSWER $2n$

7. Why does the pattern fail for a cylinder?

ANSWER Its surface is not made of flat polygons

07 Homework

Homework — 5 tasks

Check every set of three counts with $F + V - E = 2$ before writing it down.

- Count the faces, edges and vertices of a pentagonal pyramid and check the pattern.
- Do the same for an octagonal prism.
- Name three solids with a curved surface and say how many flat faces each has.
- A solid has 9 faces and 9 vertices. How many edges must it have?
- Explain in one sentence why a cuboid is a kind of prism.